

SEARCHING

PROBLEM 6

QUESTION:

DS	Session	Searching	Question Information	Level 1	Challenge 6
Problem Description: Kanna is upset to learn that no one at his school recognises his first name. Even his friends refer to him by his surname. Frustrated, he decides to make his fellow college students know his first name by forcing them to solve this question. The task is determining the third greatest number in the supplied array. Constraints: $0 < n < 100$ $0 < arr < 1000$ Input Format: first line represents the number of elements N to be get second line indicates input elements according to N Output Format: Single line represents the out put that is third largest number.					

Logical Test Cases

Test Case 1	Test Case 2
INPUT (STDIN)	INPUT (STDIN)
6 1 14 2 16 10 20	7 19 -10 20 14 2 16 10
EXPECTED OUTPUT	EXPECTED OUTPUT
The third Largest element is 14	The third Largest element is 16

Mandatory Test Cases

Test Case 1
KEYWORD
<pre>void thirdLargest(int arr[],int arr_size)</pre>

Complexity Test Cases

Test Case 1	Test Case 2	Test Case 3
CYCLOMATIC COMPLEXITY	TOKEN COUNT	NLOC
7	220	44

SOURCE CODE:

```
#include <stdio.h>
#include <limits.h>

int main() {
    int count;
    if (scanf("%d", &count) != 1) return 0;

    int nums[1005];
    for (int i = 0; i < count; i++) scanf("%d", &nums[i]);

    int max1 = INT_MIN, max2 = INT_MIN, max3 = INT_MIN;
    for (int i = 0; i < count; i++) {
        int x = nums[i];
        if (x > max1) {
            max3 = max2;
            max2 = max1;
            max1 = x;
        } else if (x > max2 && x != max1) {
            max3 = max2;
            max2 = x;
        } else if (x > max3 && x != max2 && x != max1) {
            max3 = x;
        }
    }

    printf("The third Largest element is %d\n", max3);
    return 0;
}
```

OUTPUT:

```
6
1 14 2 16 10 20
The third Largest element is 14
```

```
7
19 -10 20 14 2 16 10
The third Largest element is 16
```